

Leo Greenwood

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EDUCATION

University of Sheffield:

Sep 2024 - June 2029

MEng: Mechanical Engineering with an Industrial Placement Year ; **Currently achieving a 1st (77% average)**

Relevant Modules: Heat Transfer (90%) ; Mathematics for Engineering Modelling (84%) ; Fluids Engineering (83%) ; Computational and Numerical Methods (80%) ; Design Project Part 1 & 2 (Avg: 74%) ; Dynamics of Structures and Machines (73%)

Barnsley Sixth form College:

Sep 2022 - June 2024

A-Levels: Mathematics (A) ; Physics (A) ; Geography (B)

SKILLS

- **Design Software:** Fusion 360 ; AutoCAD ; Bambu Studio
- **Manufacturing Skills:** 3D Printing ; Hand Tools ; Laser Cutting ; Lathes ; CNC Milling ; Arduino
- **Programming Languages:** MATLAB ; Python (Pandas, SciPy, Flask)
- **Simulation Software:** Fusion 360 (FEA) ; Ansys Fluent (CFD) ; Ansys Mechanical (FEA)
- **Project Management:** Gantt Charts
- **Microsoft Office:** Word ; PowerPoint ; Excel
- **Driving Licence:** Full UK Driving Licence

ACADEMIC ENGINEERING EXPERIENCE

2nd Year Design Project:

Oct 2025 - May 2026

- **Collaborated** within a five-person team to design, manufacture, and test a functional gearbox compliant with EAPC regulations, **managing project milestones** and **negotiating design trade-offs** between performance and manufacturability.
- Performed **FEA-driven sensitivity studies** in **Fusion 360** to evaluate gear tooth stresses, applying **critical thinking** and **data-driven decision-making** to identify critical design parameters.
- **Manufactured**, assembled, and tested the prototype, utilising **iterative problem-solving** and **cross-functional communication** to refine mechanical efficiency and present findings in a detailed **technical presentation**.

Engineering: You're Hired:

Feb 2026

- **Collaborated** within a **multidisciplinary team** to develop a **cost-effective** and **sustainable engineering solution** addressing emissions within the aviation industry under a **five day project deadline**.
- Evaluated technical, environmental and commercial factors to justify design decisions, strengthening **analytical thinking** and **engineering judgement**.
- Demonstrated **strong verbal communication** through pitching final solution to industry representatives, receiving the **Dr Trish Murray Professional Behaviours Team award**.

1st Year Design Project:

Feb 2025 - May 2025

- **Collaborated** within a five-person team to design, manufacture, and test a bottle rocket, **managing team communication** and applying **fundamental engineering principles** to balance design trade-offs.
- Developed a **MATLAB** model to simulate thrust and projectile flight phases, applying **analytical thinking** to predict trajectory and inform design decisions prior to physical testing.
- Conducted launch testing and applied **data analysis** to compare experimental results against simulation predictions, utilising **iterative problem-solving** to evaluate model accuracy and present findings in a **technical review**.

PERSONAL PROJECTS

Quarter Car Suspension Simulation:

June 2026 - July 2026

- Developed a 2 DOF **state-space simulation** in **Python** and **MATLAB** to model chassis displacement and peak body acceleration, applying **analytical thinking** to evaluate sprung and unsprung mass dynamics over an 80mm speed bump.
- Implemented automated loop scripts to evaluate damping ratios, utilising **data-driven decision-making** and **trade-off optimisation** to identify the ideal balance between settling time and peak body acceleration.
- Designed a complete double wishbone suspension system (knuckle, wishbones, spring-damper unit) in **Fusion 360** using Sheffield Eco-Motorsport specifications, demonstrating **technical drawing** and **parametric CAD modelling** capabilities.
- Conducted **FEA structural studies** in **ANSYS Mechanical** to validate component integrity under load, using **iterative problem-solving** to optimise geometry and present key findings in a detailed **engineering report**.

Aerodynamic Evolution of a Car Wing Mirror:

July - August 2026

- **Reverse engineered** a 2016 Vauxhall Corsa wing mirror from physical measurements and photographs, applying **high attention to detail** to generate technical blueprints and recreate complex surface geometry in **Fusion 360** using lofts and sweeps.
- Demonstrated **strong problem-solving** and **dimensional accuracy** by resolving image lens distortion and blueprint misalignment through **physical cross-measurement**, ensuring **CAD accuracy** in Fusion 360.
- Validated the CAD model by configuring and **3D printing** a physical prototype, using **iterative problem-solving** and **cross-validation** to refine component geometry prior to aerodynamic testing.
- Conducted **CFD simulations** in **ANSYS Fluent** to compare airflow, pressure distribution, wake formation, and drag coefficients against a 1980s baseline mirror, using **critical analysis** and **data synthesis** to evaluate generational aerodynamic improvements.

WORK EXPERIENCE

Parcel Force Worldwide: Warehouse Operative

June 2026 - Current

- **Collaborated** within a fast-paced logistics team to sort high volumes of time-sensitive shipments, demonstrating **attention to detail** and **process efficiency** when routing items accurately under strict time pressure.
- Maintained **safety compliance** and hazard awareness while operating in a high-throughput warehouse environment, adhering strictly to manual handling protocols.
- Executed **workflow coordination** and **time management** alongside team members to eliminate bottlenecks and ensure continuous belt operation during peak distribution hours.
- Managed internal logistics by moving parcel cages between processing stages, relying on **proactive communication** to streamline handover operations for night shift teams and drivers.

Bright Network's IEUK 2026:

June 2026

- Completed an intensive virtual engineering experience, engaging with industry leaders to expand **commercial awareness** and evaluate core principles of **sustainable infrastructure**.
- Developed a **Python** script utilising **Pandas** to process and clean AeroGrid simulation datasets, applying **analytical thinking** and **data manipulation** techniques under tight schedules.
- Containerised data workflows using **Docker**, demonstrating **adaptability**, **self-directed learning**, and modern software deployment practices.

HOBBIES & INTERESTS

- **Motorsport:** F1 Society (member) ; watching Formula 1
- **Outdoor Pursuits:** Dog walking
- **Design & Prototyping:** 3D printing ; personal engineering projects
- **Professional:** Student Member of IMechE